

27th Jan 2026 | CS6370

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→ IR (contd.)

- What are we going to search on?
- Search engine is also an IR system.

constraints
→ Time
→ Cost
→ Space

info. originates
in my mind
(info. need.)

↓
itself gets
valued
over time.

info. gets
converted to
a query which
is admissible
by LLM.

- what we are trying to optimize? Time, but where exactly? The entire process time period is impt. to know.
- 5 documents → "tokening" → spell check, → suffix tripping → stopword removal
- concept of 'stemming' vs 'lemmatization'
- concept of classical AI vs Empirical / (old fashioned)

Bottom-up AI

→ top-down

(modern)

→ find out the
rules to comprehend tasks

→ bottom-up

→ knowledge acquisition
(solving issues)
(just a bottleneck)

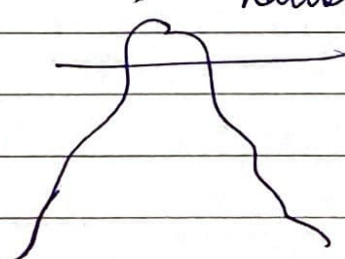
Rational intelligence

non-conscious
process
(just a
process
goes on)
worlds

emotional
conscious

physical tasks
human cognition
(can't articulate, but feasible)

(we love
mother,
but why)



- structure of classical AI, but use ML as a missing pump. That's why ML is still used as a core structure of AI systems.
- efficiency vs effectiveness
(in time) (how it solves my problem)
(& space dimension)
- Classical AI → Stop words → have many documents, eliminate those words which aren't relevant. So, we need a discriminating power. Maybe use a list of stop words. The notion of stop word depends on the domain of documents we are dealing with. Or we can assign weights to them, to discriminate them.
- Take a union → build a feature space → then develop a vector for each document.
- IDF (Inverse document freq.) → $\log_2\left(\frac{N}{n}\right)$

$\log_2\left(\frac{1}{P}\right)$
 $\downarrow$
 Probab. eq. $\text{idf}(\text{"the"}) = \log_2\left(\frac{N}{n_{\text{the}}}\right)$ $\rightarrow$ it helps in scaling.

$\downarrow$ if this is closer to 0. That means this word appears in all these documents. Therefore, this word does not have any discriminating power.

if P is small

$$IDF = \log_2 \left(\frac{N}{n} \right) = \log_2 \left(\frac{1}{\frac{n}{N}} \right)$$

info. content ← relative freq. becomes P ←

- Symbolic vs Empirical AI (material)